NANDHA PRAKASH

Assistant Software Engineer

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PROFESSIONAL SUMMARY

Trainee Engineer with hands-on experience in Python development, backend systems, PostgreSQL, Docker, and AI-driven document processing applications. Skilled in building data pipelines, API integrations, and scalable automation solutions, with a strong interest in software engineering, machine learning, and system design.

EDUCATION

UG[B-TECH (INFORMATION TECHNOLOGY)] **2021 - 2025**
SONA College of Technology ,Salem - 8.6 cgpa

HSC **2020 - 2021**
SRK Matriculation Higher Secondary school, Salem - 89.25%

SSLC **2018 - 2019**
The Little Flower Matriculation school, Salem -89.8%

TECHNICAL SKILL

- **Programming Languages:** Python, SQL, Typescript,Nodejs
- **AI & Machine Learning:** Large Language Models (LLMs), RAG,Machine Learning , Prompt Engineering
- **Frameworks & Libraries:** FastAPI, Pandas, SQLAlchemy, Boto3, pdf2image, Regex, PyTorch, PyPDF2
- **Databases & Search Technologies:** PostgreSQL, Neo4j ,OpenSearch Database Design
- **Backend & Architecture:** RESTful APIs, API Integration, Microservices Architecture
- **DevOps & Infrastructure:** Docker, Kubernetes (Basic)
- **Tools & Platforms:** Git, GitHub, Claude Code, Antigravity

PROFESSIONAL EXPERIENCE

TRAINEE ENGINEER - SLATE AI | BANGALORE,INDIA **JUN 2025- PRESENT**

- Developed backend applications using Python, FastAPI, and PostgreSQL.
- Built and integrated RESTful APIs for business workflows.
- Developed AI-powered backend services and data processing pipelines.
- Designed database schemas and optimized SQL queries.
- Collaborated with teams to resolve production issues and deliver solutions.
- Containerized applications using Docker and supported deployments.

INTERN - SLATE AI | BANGALORE,INDIA **FEB 2025- MAY 2025**

- Developed AI-based document parsing and data extraction pipelines for PDF documents.
- Built API validation workflows with custom error handling.
- Wrote and optimized PostgreSQL queries for data retrieval and reporting.
- Developed and executed test cases for APIs and backend modules.
- Collaborated with cross-functional teams in an Agile environment.
- Utilized Git, Postman, Jira, and database tools for development and testing.

PROJECTS

Smart Search Platform – Intelligent AI Search Engine

Tech Stack: Python, FastAPI, Neo4j, OpenSearch, pgvector, Docker, OpenAI API

- Developed an AI data ingestion pipeline to parse complex documents for multi-database retrieval.
- Implemented a hybrid search architecture using pgvector, Neo4j , and OpenSearch.
- Built high-performance, containerized REST APIs in Python integrating LLMs to dynamically generate and execute SQL and database queries from natural language.
- Integrated OpenAI language models to power advanced NLP and contextual search capabilities.

Site Planning – AI-Driven Layout Optimizer

Tech Stack: Python, FastAPI, OpenAI API, Shapely, GeoPandas, PostgreSQL, WebSockets

- Developed an AI-powered spatial optimization engine for automated site layout generation.
- Integrated Shapely, GeoPandas, and OpenAI models to generate optimized 2D and 3D site plans.
- Built FastAPI and WebSocket services for real-time layout processing and visualization.
- Automated report generation and site visualizations using Jinja2 and Playwright.

ACHIEVEMENTS

- Qualified for the finals, ranked in the top 7 at the Smart Indian Hackathon (SIH)
- Organized LEEF events for the English Club at Sona College of Technology.